AI-AUGMENTED METHODS FOR INTERPRETABLE AND EFFICIENT MODELING IN MODERN ASTRONOMY

25-26J-108

SLIIT
FACULTY OF COMPUTING

INTERPRETABLE GALAXY FORMATION MODELING VIA PHYSICS-INFORMED NEURAL NETWORKS (PINNS) AND EXPLAINABLE AI (XAI)

Specialization: Data Science

Interpretable Galaxy Formation Modeling via Physics-Informed Neural Networks (PINNs) and Explainable AI (XAI)

Knowledge Gap

- Can PINNs, combined with explainability tools, offer interpretable and physically consistent simulations of galaxy formation?

Shortcomings of Existing Systems

- Lack of physical constraints in existing models
- Limited ability to explain predictions
- Heavy reliance on simulated data

Interpretable Galaxy Formation Modeling via Physics-Informed Neural Networks (PINNs) and Explainable AI (XAI)

Application of Data Science Domains

- Physics Informed Neural Networks (PINNs)
- Explainable AI (XAI)

Technologies Used

- Astropy
- SHAP
- PyTorch

Interpretable Galaxy Formation Modeling via Physics-Informed Neural Networks (PINNs) and Explainable AI (XAI)

Evaluation Methods

Comparing against "black box" models / RMSE, R^2 metrics

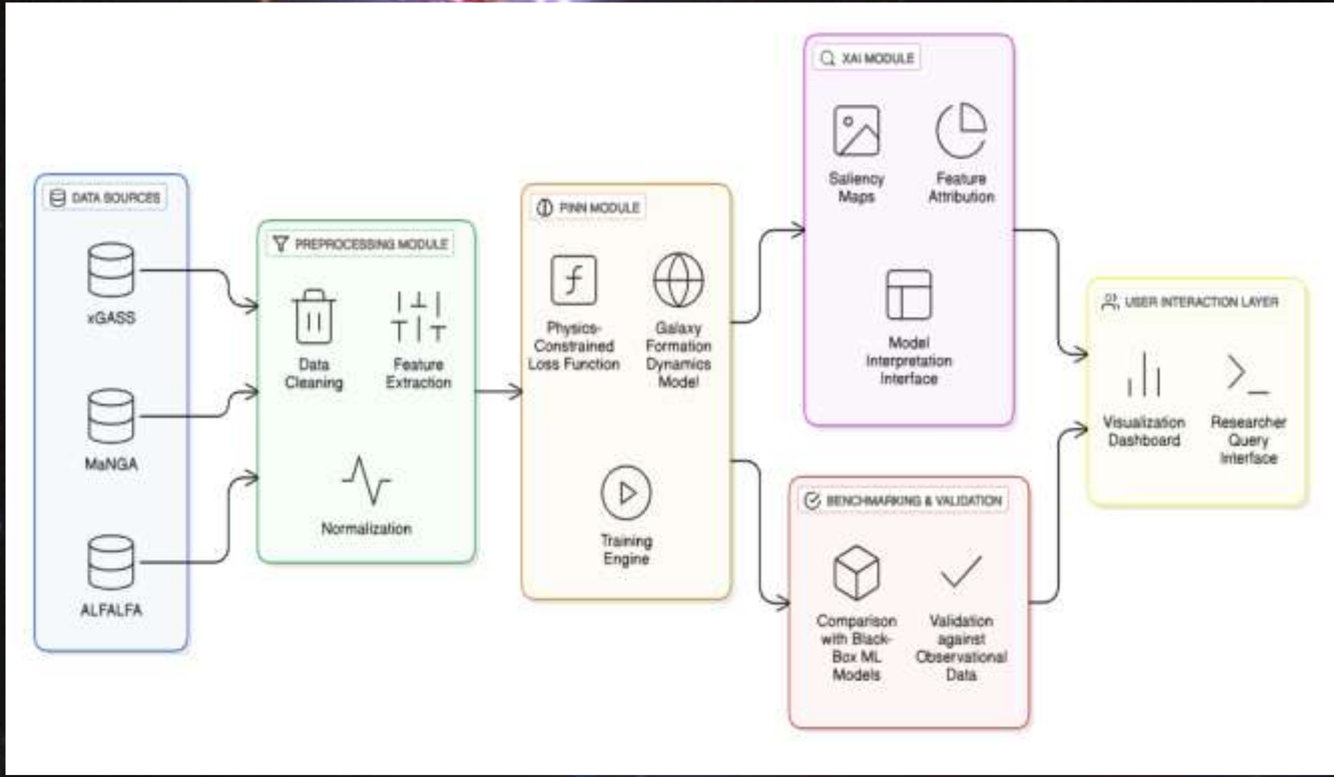
Improved accuracy over the "black box" models

Data Availability

Publicly accessible astrophysics data (e.g.: MaNGA, ALFALFA, xGASS, etc)

Open-source/can be used for researches

Interpretable Galaxy Formation Modeling via Physics-Informed Neural Networks (PINNs) and Explainable AI (XAI)



- Uncovering hidden galaxy formation pathways / dark matter.
- Building trust in machine learning for science.
- Transparent physics respecting “cosmic-sandbox”.
- Guiding next generation telescope observations & astrophysics research.

CONTRASTIVE LEARNING FOR CLASSIFICATION OF RARE ASTRONOMICAL TRANSIENTS

Specialization: Data Science

CONTRASTIVE LEARNING FOR CLASSIFICATION OF RARE ASTRONOMICAL TRANSIENTS

Knowledge Gap	Shortcomings of Existing Systems
• How can we improve the classification of rare astronomical transients using contrastive learning on light curve data?	• Needs big labelled dataset • Poor rare event detection • High Computational Cost

CONTRASTIVE LEARNING FOR CLASSIFICATION OF RARE ASTRONOMICAL TRANSIENTS

- **Application of Data Science Domains**
  - Contrastive Learning
  - Deep Learning
- **Technologies Used**
  - SimCLR
  - Astropy
  - PyTorch

CONTRASTIVE LEARNING FOR CLASSIFICATION OF RARE ASTRONOMICAL TRANSIENTS

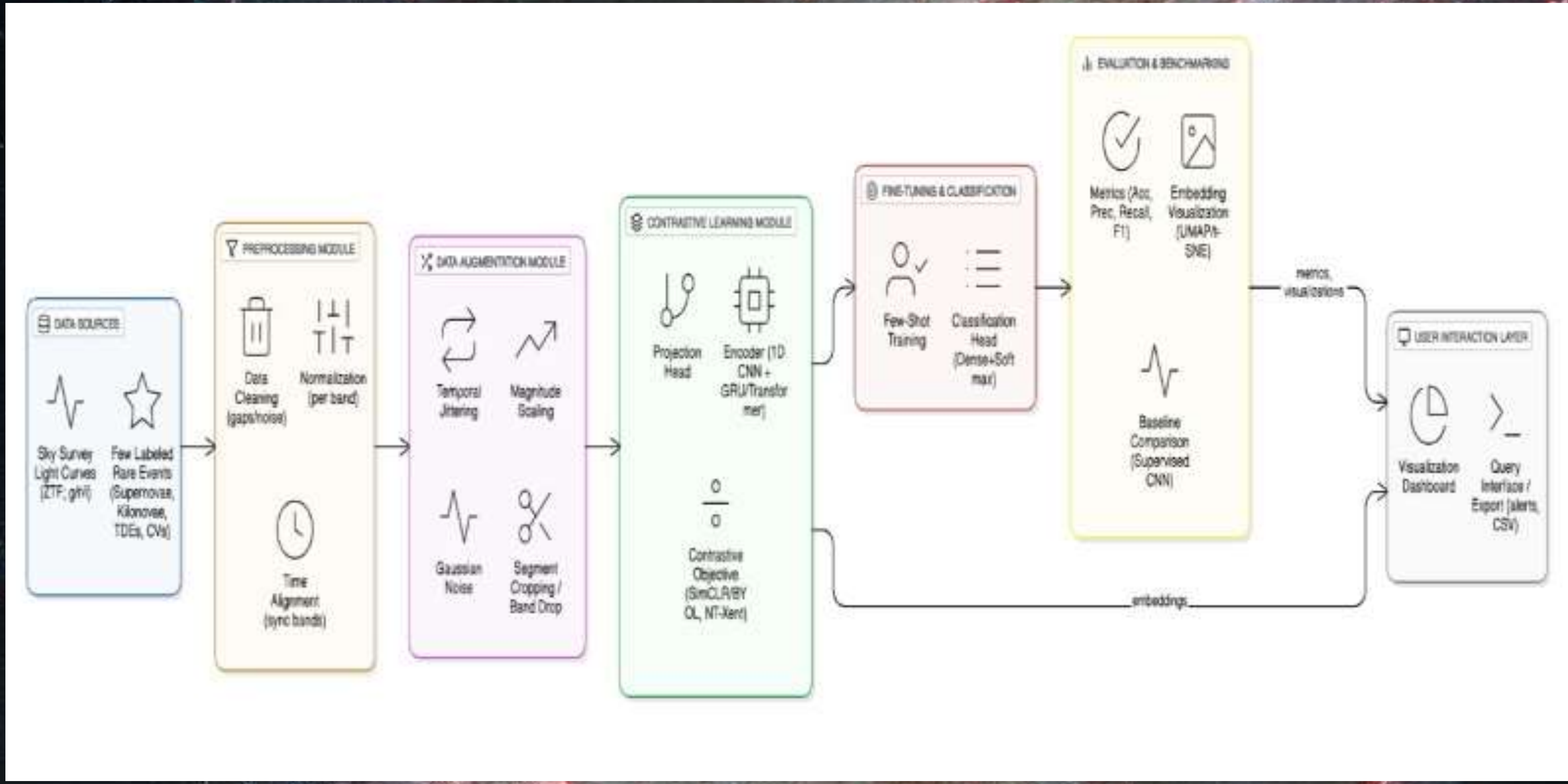
Evaluation Methods

- Use metrics such as Accuracy, Precision, Recall, and F1-score.
- Compare against baseline supervised models trained only on labelled data

Data Availability

- ZTF (Zwicky Transient Facility) multi-band light curve data
- Public datasets, no ethical clearance needed

CONTRASTIVE LEARNING FOR CLASSIFICATION OF RARE ASTRONOMICAL TRANSIENTS



- Can be used in big telescope surveys to automatically check nightly data and find rare events.
- Helps astronomers find rare space events (like supernovae)faster.
- Works well even when only a few labelled examples are available.
- Supports scientists in studying new or unusual cosmic events.

STELLAR POPULATION CHARACTERIZATION VIA SELF-SUPERVISED LEARNING

Specialization: Data Science

STELLAR POPULATION CHARACTERIZATION VIA SELF-SUPERVISED LEARNING

Knowledge Gap

- How can we extract meaningful representations of stellar populations in distant, low-signal galaxies without relying on labeled data?

Shortcomings of Existing Systems

- High dependency on costly telescope data and labeled datasets.
- Low-resolution/incomplete data.

STELLAR POPULATION CHARACTERIZATION VIA SELF-SUPERVISED LEARNING

- Application of Data Science Domains
  - Self – Supervised Learning
  - Feature Engineering

- Technologies Used
  - Astropy
  - PyTorch
  - SciPy

STELLAR POPULATION CHARACTERIZATION VIA SELF-SUPERVISED LEARNING

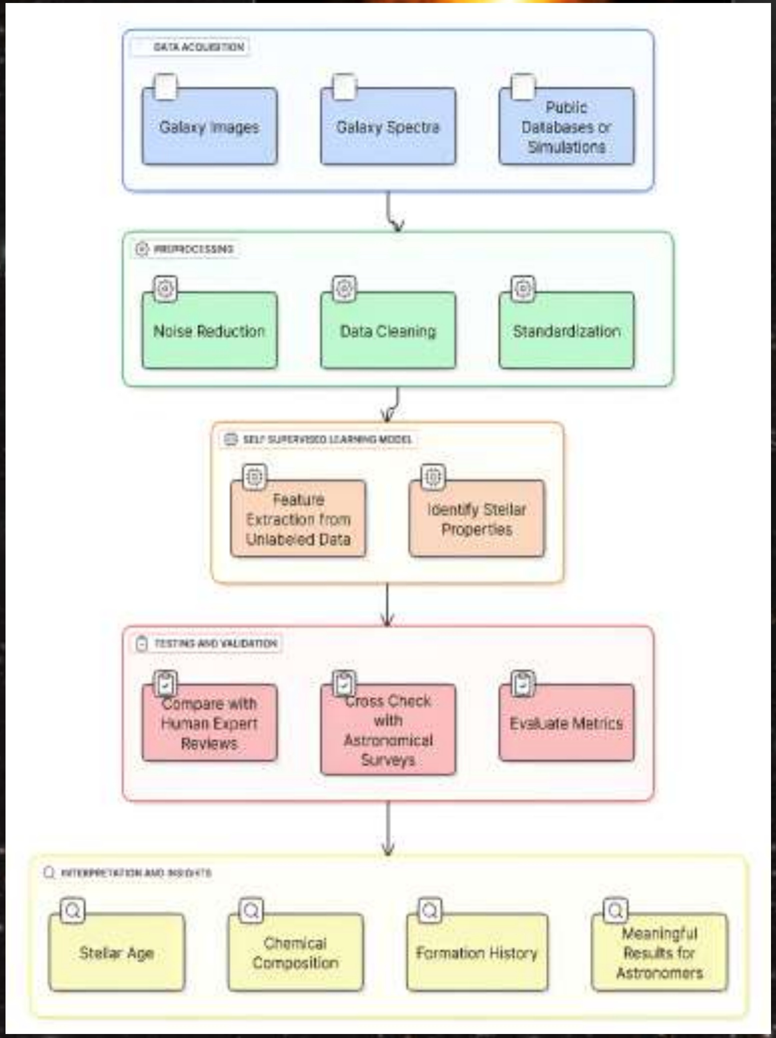
Evaluation Methods

- Compare with Human Expert Reviews.
- Use Performance Metrics

Data Availability

- Publicly accessible data repositories (e.g., SDSS, Hubble, JWST)
- Open-source datasets

STELLAR POPULATION CHARACTERIZATION VIA SELF-SUPERVISED LEARNING



- Analyze faint and distant galaxies without large labeled datasets.
- Accelerate discoveries in stellar evolution, galaxy formation, and cosmic history.
- Helps astronomers make better use of low-resolution or incomplete survey data
- Provides insights into star formation, chemical enrichment, and galaxy evolution

QUASAR VARIABILITY MODELING USING ATTENTION-AUGMENTED RNNS

Specialization: Data Science

QUASAR VARIABILITY MODELING USING ATTENTION-AUGMENTED RNNS

Knowledge Gap

- Can attention-based models identify the most important time intervals in a quasar’s light curve to explain its brightness variability?

Shortcomings of Existing Systems

- Traditional spectroscopy
- Deep learning without attention
- Failure to capture complex variability

QUASAR VARIABILITY MODELING USING ATTENTION-AUGMENTED RNNS

- **Application of Data Science Domains**
  - Attention Augmented RNN
  - Time Series Analysis
- **Technologies Used**
  - Astropy
  - SciPy
  - PyTorch

QUASAR VARIABILITY MODELING USING ATTENTION-AUGMENTED RNNS

Evaluation Methods

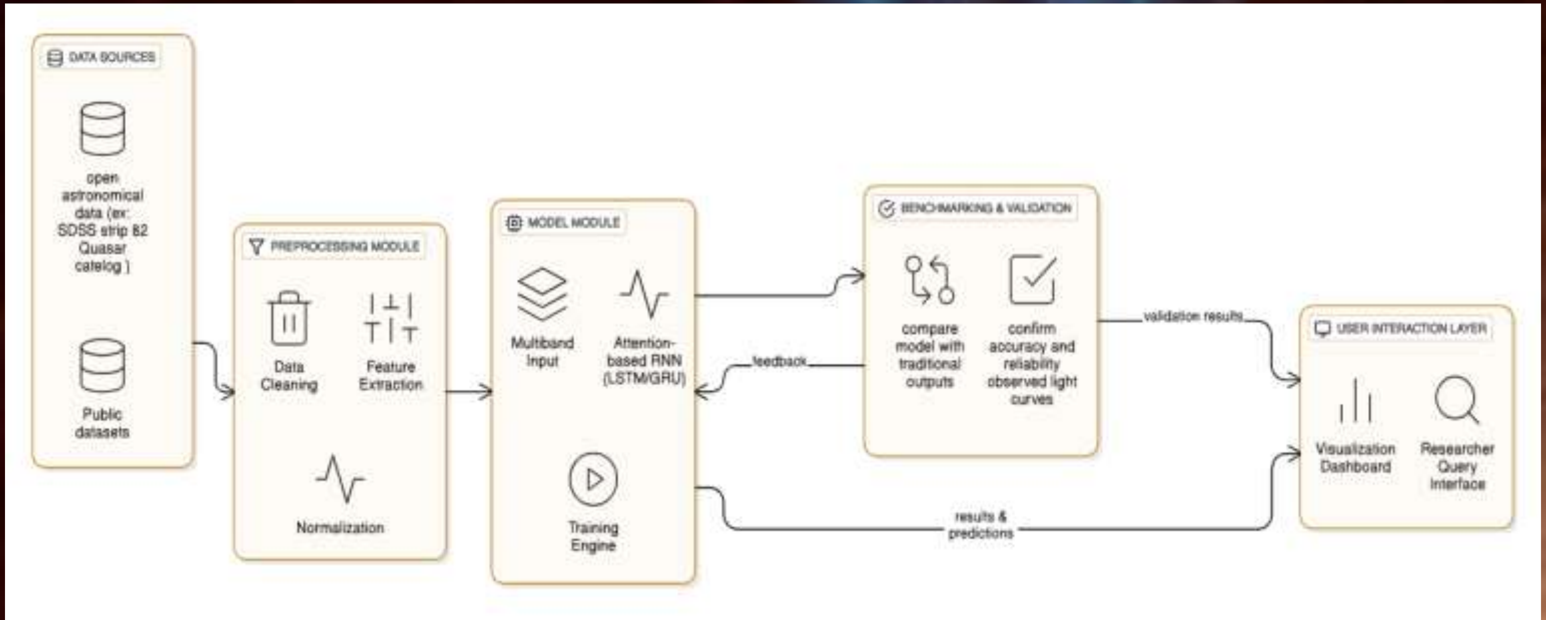
- Compare model predictions with traditional outputs
- Confirm accuracy and reliability of the observed light curves

Data Availability

- Open astronomical data (Ex : SDSS Stripe 82 Quasar Catalog

  Pan-STARRS DR2)
- Public datasets

QUASAR VARIABILITY MODELING USING ATTENTION-AUGMENTED RNNS



- Use brightness over time to spot luminosity Fluctuation of quasars
- Capture complex quasar variability with attention-augmented RNNs
- Compare model predictions with traditional outputs
- Derive Insights about Black Hole Properties

QUASAR VARIABILITY MODELING USING ATTENTION-AUGMENTED RNNS

Q&A

THANK YOU!